

Assignment 5b

Sunday, 31 March 2024 3:41 pm

Assignment 5b

(not Q1)

(not Q2)

Q3 already proved in class (i.e. the C-S inequality) but you should also read the proof of Theorem 6.2.1 in the book

Do Q4

(not Q5) instead we do Exercise set 6.2, Q6

Do Q6

For Question no. 7, do the same numbered questions from Exercise set 6.3 (i.e. 17 (a), 18, 25, 26, 27, 29

(not Q8)

[Refer to Lecture 21 slides, pages 8-9 for details of the questions included]

(2) PROVE CAUCHY-SCHWARZ INEQUALITY IN CASE OF EUCLIDEAN INNER PRODUCT.
HINT: SEE P. 300 7TH ED.
OR P. 287 8TH ED.
(3) ALSO PROVE (2) IN GENERAL.

See book

(4) IF u AND v ARE TWO VECTORS IN AN INNER PRODUCT SPACE THEN
(i) $\|u+v\| \leq \|u\| + \|v\|$ } TRIANGLE INEQUALITY
(ii) USING THE FACT THAT u AND v ARE ORTHOGONAL PROVE THAT $\|u+v\|^2 = \|u\|^2 + \|v\|^2$
WHICH IS ALSO A SPECIAL CASE OF THE TRIANGLE INEQUALITY

PROVE THAT $\|u+v\|^2 = \|u\|^2 + \|v\|^2$
 WHICH IS ALSO CALLED A
 GENERALIZED THEOREM OF
 PYTHAGORAS.

See book for (i) & (ii) Home work (previous)

6. Let P_2 have the inner product in Example 8 of Section 6.1. Find the cosine of the angle between p and q .

(a) $p = -1 + 5x + 2x^2, q = 2 + 4x - 9x^2$

(b) $p = x - x^2, q = 7 + 3x + 3x^2$

Question 06(a)

Solution: $p = -1 + 5x + 2x^2, q = 2 + 4x - 9x^2$

$$\theta = \cos^{-1} \left(\frac{p \cdot q}{\|p\| \|q\|} \right) = \cos^{-1} \left(\frac{-1 \cdot 2 + 5 \cdot 4 + 2 \cdot (-9)}{\sqrt{(-1)^2 + 5^2 + 2^2} \sqrt{2^2 + 4^2 + (-9)^2}} \right)$$

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(6) LET $f(x)$ AND $g(x)$ BE
 CONTINUOUS FUNCTIONS ON $[0,1]$
 PROVE $\left[\int_0^1 f(x)g(x) dx \right]^2$
 $\leq \left[\int_0^1 f^2(x) dx \right] \left[\int_0^1 g^2(x) dx \right]$

Proof

$\forall x \in \mathbb{R} : 0 \leq (xf(t) + g(t))^2 \rightarrow$ Main Assumption

$$0 \leq \int_a^b (xf(t) + g(t))^2 dt$$

Relative Sizes of Definite Integrals

$$= x^2 \int_a^b (f(t))^2 dt + 2x \int_a^b f(t) g(t) dt + \int_a^b (g(t))^2 dt$$

Linear Combination of Definite Integrals

$$= Ax^2 + 2Bx + C$$

where:

$$A = \int_a^b (f(t))^2 dt$$

$$B = \int_a^b f(t) g(t) dt$$

$$C = \int_a^b (g(t))^2 dt$$

since these integral would be scalar

punch line

The quadratic equation $Ax^2 + 2Bx + C$ is non-negative for all x .

It follows (using the same reasoning as in Cauchy's Inequality) that the discriminant $(2B)^2 - 4AC$ of this polynomial must be non-positive.

Thus:

$$B^2 \leq AC$$

and hence the result.

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(7) GRAM-SCHMIDT PROCESS AND ORTHONORMALITY

17(a), 18, 25, 26, 27, 29

- Let $\mathbb{R}^3$ have the Euclidean inner product. Use the Gram-Schmidt process to transform the basis $\{u_1, u_2, u_3\}$ into an orthonormal basis.

(a) $u_1 = (1, 1, 1), u_2 = (-1, 1, 0), u_3 = (1, 2, 1)$

$$u_1 \rightarrow v_1 = \frac{u_1}{\|u_1\|}$$

$$v_2 = u_2 - \langle u_2, v_1 \rangle v_1$$

17. (a) Let

$$v_1 = \frac{u_1}{\|u_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \cdot (-1, 1, 0) = 0$$

$$v_2 = u_2$$

Since $\langle \mathbf{u}_2, \mathbf{v}_1 \rangle = 0$, we have

$$\mathbf{v}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \rightarrow$$

Since $\langle \mathbf{u}_3, \mathbf{v}_1 \rangle = \frac{4}{\sqrt{3}}$ and $\langle \mathbf{u}_3, \mathbf{v}_2 \rangle = \frac{1}{\sqrt{2}}$, we have

$$\begin{aligned} \mathbf{u}_3 - \langle \mathbf{u}_3, \mathbf{v}_1 \rangle \mathbf{v}_1 - \langle \mathbf{u}_3, \mathbf{v}_2 \rangle \mathbf{v}_2 \\ = (1, 2, 1) - \frac{4}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) - \frac{1}{\sqrt{2}} \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) \end{aligned}$$

$$= \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3} \right) \leftarrow \mathcal{N}$$

This vector has norm $\left\| \left(\frac{1}{6}, \frac{1}{6}, -\frac{1}{3} \right) \right\| = \frac{1}{\sqrt{6}}$. Thus

$$\mathbf{v}_3 = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right)$$

and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is the desired orthonormal basis.

Let $\mathbb{R}^4$ have the Euclidean inner product. Use the Gram-Schmidt process to transform the basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ into an orthonormal basis.

$$\mathbf{u}_1 = (0, 2, 1, 0), \quad \mathbf{u}_2 = (1, -1, 0, 0), \quad \mathbf{u}_3 = (1, 2, 0, -1), \quad \mathbf{u}_4 = (1, 0, 0, 1)$$

$$\bar{\mathbf{v}}_1 = \frac{\bar{\mathbf{u}}_1}{\|\bar{\mathbf{u}}_1\|}, \quad \mathbf{v}_2 = \frac{\bar{\mathbf{u}}_2^*}{\|\bar{\mathbf{u}}_2^*\|} \quad \therefore \bar{\mathbf{u}}_2^* = \bar{\mathbf{u}}_2 - \langle \bar{\mathbf{u}}_2, \bar{\mathbf{v}}_1 \rangle \bar{\mathbf{v}}_1$$

$$\mathbf{v}_3 = \frac{\bar{\mathbf{u}}_3^*}{\|\bar{\mathbf{u}}_3^*\|} \quad \therefore \bar{\mathbf{u}}_3^* = \bar{\mathbf{u}}_3 - \langle \bar{\mathbf{u}}_3, \bar{\mathbf{v}}_1 \rangle \bar{\mathbf{v}}_1 - \langle \bar{\mathbf{u}}_3, \bar{\mathbf{v}}_2 \rangle \bar{\mathbf{v}}_2$$

$$\mathbf{v}_4 = \frac{\bar{\mathbf{u}}_4^*}{\|\bar{\mathbf{u}}_4^*\|} \quad \therefore \bar{\mathbf{u}}_4^* = \bar{\mathbf{u}}_4 - \sum_{i=1}^3 \langle \bar{\mathbf{u}}_4, \bar{\mathbf{v}}_i \rangle \bar{\mathbf{v}}_i$$

Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be an orthonormal basis for an inner product space V . Show that if $\mathbf{w}$ is a vector in V , then

$$25. \quad \|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{v}_1 \rangle^2 + \langle \mathbf{w}, \mathbf{v}_2 \rangle^2 + \langle \mathbf{w}, \mathbf{v}_3 \rangle^2. \quad \therefore \|\mathbf{w}\|^2 = \omega_1^2 + \omega_2^2 + \omega_3^2 + \dots + \omega_n^2$$

25. By Theorem 6.3.1, we know that

$$\mathbf{w} = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + a_3 \mathbf{v}_3$$

where $a_i = \langle \mathbf{w}, \mathbf{v}_i \rangle$. Thus

$$\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w} \rangle$$

$$\begin{aligned} &= a_1^2 (\mathbf{v}_1, \mathbf{v}_1) + a_1 a_2 (\mathbf{v}_1, \mathbf{v}_2) + a_1 a_3 (\mathbf{v}_1, \mathbf{v}_3) \\ &= a_1^2 + a_2^2 + a_3^2 \end{aligned}$$

25. By Theorem 6.3.1, we know that

$$\mathbf{w} = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + a_3 \mathbf{v}_3$$

where $a_i = \langle \mathbf{w}, \mathbf{v}_i \rangle$. Thus

$$\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{w} \rangle$$

$$= \sum_{i=1}^3 a_i^2 \langle \mathbf{v}_i, \mathbf{v}_i \rangle + \sum_{i \neq j} a_i a_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle$$

But $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ if $i \neq j$ and $\langle \mathbf{v}_i, \mathbf{v}_i \rangle = 1$ because the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is orthonormal. Hence

$$\|\mathbf{w}\|^2 = a_1^2 + a_2^2 + a_3^2$$

$$= \underbrace{\langle \mathbf{w}, \mathbf{v}_1 \rangle^2}_{a_1^2} + \underbrace{\langle \mathbf{w}, \mathbf{v}_2 \rangle^2}_{a_2^2} + \underbrace{\langle \mathbf{w}, \mathbf{v}_3 \rangle^2}_{a_3^2}$$

$$(a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + a_3 \mathbf{v}_3) \cdot \mathbf{w}$$

26. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an orthonormal basis for an inner product space V . Show that if $\mathbf{w}$ is a vector in V , then $\|\mathbf{w}\|^2 = \langle \mathbf{w}, \mathbf{v}_1 \rangle^2 + \langle \mathbf{w}, \mathbf{v}_2 \rangle^2 + \dots + \langle \mathbf{w}, \mathbf{v}_n \rangle^2$.

27. In Step 3 of the proof of Theorem 6.3.6, it was stated that "the linear independence of $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ ensures that $\mathbf{v}_3 \neq \mathbf{0}$." Prove this statement.

Base

27. Suppose the contrary; that is, suppose that

(*)

$$\mathbf{u}_3 - \langle \mathbf{u}_3, \mathbf{v}_1 \rangle \mathbf{v}_1 - \langle \mathbf{u}_3, \mathbf{v}_2 \rangle \mathbf{v}_2 = \mathbf{0} \Rightarrow \bar{u}_3 = \underbrace{\langle \bar{u}_3, \bar{v}_1 \rangle}_{\text{scalar}} \bar{v}_1 + \underbrace{\langle \bar{u}_3, \bar{v}_2 \rangle}_{\text{scalar}} \bar{v}_2$$

Then (*) implies that $\mathbf{u}_3$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. But $\mathbf{v}_1$ is a multiple of $\mathbf{u}_1$ while $\mathbf{v}_2$ is a linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$. Hence, (*) implies that $\mathbf{u}_3$ is a linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$ and therefore that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is linearly dependent, contrary to the hypothesis that $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is linearly independent. Thus, the assumption that (*) holds leads to a contradiction.

29. (For Readers Who Have Studied Calculus)

Let the vector space P_2 have the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^1 p(x)q(x)dx$$

Apply the Gram-Schmidt process to transform the standard basis $\mathcal{S} = \{1, x, x^2\}$ into an orthonormal basis. (The polynomials in the resulting basis are called the first three *normalized Legendre polynomials*.)

29. We have $\mathbf{u}_1 = 1$, $\mathbf{u}_2 = x$, and $\mathbf{u}_3 = x^2$. Since

$$\langle \mathbf{u}_1, \mathbf{u}_1 \rangle = \int_{-1}^1 1 \times 1 dx = 2$$

$$\|\mathbf{u}_1\|^2 = \langle \mathbf{u}_1, \mathbf{u}_1 \rangle = \int_{-1}^1 1 dx = 2$$

we let

$$\bar{\mathbf{v}}_1 = \frac{1}{\sqrt{2}}$$

$$\bar{\mathbf{v}}_2 = \bar{\mathbf{u}}_2 - \langle \bar{\mathbf{u}}_2, \bar{\mathbf{v}}_1 \rangle \bar{\mathbf{v}}_1 = x - \left(\int_{-1}^1 x \left(\frac{1}{\sqrt{2}} \right) dx \right) \frac{1}{\sqrt{2}}$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}$$

Then

$$\langle \mathbf{u}_2, \mathbf{v}_1 \rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 x \, dx = 0$$

and thus $\mathbf{v}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}$ where

$$\|\mathbf{u}_2\|^2 = \int_{-1}^1 x^2 \, dx = \frac{2}{3}$$

Hence

$$\mathbf{v}_2 = \sqrt{\frac{3}{2}} x$$

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In order to compute $\mathbf{v}_3$, we note that

$$\langle \mathbf{u}_3, \mathbf{v}_1 \rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 x^2 \, dx = \frac{\sqrt{2}}{3}$$

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and

$$\langle \mathbf{u}_3, \mathbf{v}_2 \rangle = \sqrt{\frac{3}{2}} \int_{-1}^1 x^3 \, dx = 0$$

Thus

$$\mathbf{u}_3 - \langle \mathbf{u}_3, \mathbf{v}_1 \rangle \mathbf{v}_1 - \langle \mathbf{u}_3, \mathbf{v}_2 \rangle \mathbf{v}_2 = x^2 - \frac{1}{3}$$

and

$$\left\| x^2 - \frac{1}{3} \right\|^2 = \int_{-1}^1 \left[x^2 - \frac{1}{3} \right]^2 \, dx = \frac{8}{45}$$

Hence,

$$\mathbf{v}_3 = \sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3} \right) \quad \text{or} \quad \mathbf{v}_3 = \frac{\sqrt{5}}{2\sqrt{2}} (3x^2 - 1)$$

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$$= x - \int_{-1}^1 \left(x \left(\frac{1}{\sqrt{2}} \right) \right) dx \cdot \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} \int_{-1}^1 x \, dx = \frac{x^2}{2} \Big|_{-1}^1 = 0$$